

KILLARA HIGH SCHOOL



TERM 2 2003

EXTENSION 2 MATHEMATICS EXAMINATION

YEAR 12

Instructions:

Time Allowed: 70 minutes

Approved calculators may be used.

Show ALL working clearly.

Marks may not be awarded for careless or badly arranged work.

Begin each question on a new page.

Question 1

Marks

- i) Use integration by parts to evaluate $\int_1^3 x^2 \ln x \, dx$ 2
- ii) Evaluate $\int_3^4 \frac{3x}{2+3x} dx$ 2
- iii) Find $\int_0^1 \frac{1}{x^2+4x+5} dx$ 2
- iv) (a) If $I_n = \int \cot^n \theta d\theta$ show that $I_n = \frac{-1}{n-1} \cot^{n-1} \theta - I_{n-2}$ 3
- (b) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^4 \theta d\theta$ 2
- v) Use the substitution $x = a - t$ where a is a constant, to prove that $\int_0^a f(x) dx = \int_0^a f(a-t) dt$
- Hence, or otherwise, show that $\int_0^1 x(1-x)^{99} dx = \frac{1}{10100}$ 4

Question 2

- i) The equation $8x^3 - 54x + m = 0$ has a double root. Find the possible values of m . 3
- ii) Let α, β, γ be the roots of the equation $x^3 + qx + r = 0$ where $r \neq 0$. Obtain as functions of q, r in their simplest forms, the co-efficients of the cubic equations whose roots are:
- (a) $\alpha^2, \beta^2, \gamma^2$ 2
- (b) $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$ 2
- (c) $\alpha^{-2}, \beta^{-2}, \gamma^{-2}$ 2

Question 3

- a) Prove using mathematical induction that for $n \geq 1$
- $$1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{n^2} \leq 2 - \frac{1}{n}$$
- 4
- b) Hence prove that $1.45 \leq 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{99^2} \leq 1.99$ 2

$$i) \int_1^3 x^2 \ln x \, dx \quad 15$$

$$= \int_1^3 \ln x \, \frac{d}{dx} \left(\frac{x^3}{3} \right) dx$$

$$= \left[\frac{x^3}{3} \ln x \right]_1^3 - \int_1^3 \frac{x^3}{3} \times \frac{1}{x} dx$$

$$= \left[\frac{x^3}{3} \ln x \right]_1^3 - \int_1^3 \frac{x^2}{3} dx$$

$$= \left[\frac{x^3}{3} \ln x - \frac{x^3}{9} \right]_1^3$$

$$= 9 \ln 3 - 3 - \left(0 - \frac{1}{9} \right)$$

$$= 9 \ln 3 - 2 \frac{8}{9}$$

$$ii) \int_3^4 \frac{3x}{2+3x} dx$$

$$= \int_3^4 \left(\frac{2+3x}{2+3x} - \frac{2}{2+3x} \right) dx$$

$$= \int_3^4 \left(1 - \frac{2}{2+3x} \right) dx$$

$$= \left[x - \frac{2}{3} \ln |2+3x| \right]_3^4$$

$$= \left(4 - \frac{2}{3} \ln 14 \right) - \left(3 - \frac{2}{3} \ln 11 \right)$$

$$= 1 - \frac{2}{3} \ln \left(\frac{14}{11} \right)$$

$$iii) \int_0^1 \frac{1}{x^2 + 4x + 5} dx$$

$$= \int_0^1 \frac{1}{(x+2)^2 + 1} dx$$

$$= \left[\tan^{-1}(x+2) \right]_0^1$$

$$= \tan^{-1} 3 - \tan^{-1} 2$$

$$= (0.142 \text{ 3 dp})$$

$$iv) a) I_n = \int \cot^n \theta \, d\theta$$

$$= \int \cot^{n-2} \theta \cot^2 \theta \, d\theta$$

$$= \int \cot^{n-2} \theta (\csc^2 \theta - 1) \, d\theta$$

$$= \int \cot^{n-2} \theta \csc^2 \theta \, d\theta - I_{n-2}$$

$$= \frac{-1}{n-1} \cot^{n-1} \theta - I_{n-2}$$

$$b) \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^4 \theta \, d\theta = \left[-\frac{1}{3} \cot^3 \theta - \frac{1}{2} \cot \theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \left[-\frac{1}{3} (0^3 - 1^3) - \left(1 - \frac{1}{2} \right) \right]$$

$$= \frac{1}{3} - 1 + \frac{1}{2} = \frac{\pi}{4} - \frac{2}{3}$$

$$\text{Since } I_2 = \left[-\frac{1}{1} \cot \theta \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} - I_0$$

$$= [0 + 1] - I_0$$

$$I_0 = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta$$

$$= \frac{\pi}{2} - \frac{\pi}{4}$$

$$= \frac{\pi}{4}$$

$$1) \quad x = a - t \\ dx = -dt$$

$$\text{At } x=0 \quad t=a \\ x=a \quad t=0$$

$$\therefore \int_0^a f(x) dx = \int_a^0 f(a-t) (-dt) \\ = - \int_a^0 f(a-t) dt \\ = \int_0^a f(a-t) dt$$

$$\therefore \text{if } a=1 \text{ and } x=1-t \\ \text{and } f(x) = x(1-x)^{99} \\ \text{then } f(1-t) = (1-t)(t)^{99}$$

$$\therefore \int_0^1 x(1-x)^{99} = \int_0^1 (1-t)t^{99} dt \\ = \int_0^1 (t^{99} - t^{100}) dt \\ = \left[\frac{t^{100}}{100} - \frac{t^{101}}{101} \right]_0^1 \\ = \frac{1}{100} - \frac{1}{101} \\ = \frac{1}{10100} \quad |$$

$$2/1) \quad P(x) = 8x^3 - 54x + m$$

$$P(x) = 0$$

$$P'(x) = 0 \quad \text{since double root}$$

$$P'(x) = 24x^2 - 54$$

$$P'(x) = 24x^2 - 54 = 0$$

$$x^2 = \frac{54}{24}$$

$$= \frac{27}{12}$$

$$= \frac{9}{4}$$

$$x = \pm \frac{3}{2}$$

$$P(x) = 0$$

$$\therefore 8x^3 - 54x + m = 0$$

$$m = 54x - 8x^3$$

$$= (54 - 8x^2)x$$

$$= (54 - 8 \times \frac{9}{4}) \left(\pm \frac{3}{2} \right)$$

$$= (54 - 18) \times \pm \frac{3}{2}$$

$$= 36 \times \pm \frac{3}{2}$$

$$= \pm 54 \quad |$$

a) Let $y = x^2$ where x satisfies the equation

$$x^3 + qx + r = 0$$

then $x(x^2 + q) = -r$

$$\therefore x^2(x^2 + q)^2 = r^2$$

i.e. $y(y + q)^2 = r^2$

i.e. $y^3 + 2qy + q^2y - r^2 = 0$

thus if $x^3 + qx + r = 0$ has roots α, β, γ then

$y^3 + 2qy + q^2y - r^2 = 0$ has roots $\alpha^2, \beta^2, \gamma^2$

$\therefore$ coeffs are $1, 2q, q^2, -r^2$

or $\alpha + \beta + \gamma = 0$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

$$\alpha\beta\gamma = -r$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = -2q$$

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$$

$$(\alpha\beta + \beta\gamma + \gamma\alpha)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)$$

$$= q^2$$

$$(\alpha\beta\gamma)^2 = (-r)^2 = r^2$$

$\therefore$ eqn satisfied by $\alpha^2, \beta^2, \gamma^2$

is $y^3 + 2qy + q^2y - r^2 = 0$

b) Let $y = x^{-1}$ where x satisfies the equation $x^3 + qx + r = 0$

then $\frac{1}{y^3} + q\frac{1}{y} + r = 0$

i.e. $1 + qy^2 + ry^3 = 0$

$$\therefore y^3 + \frac{q}{r}y^2 + \frac{1}{r} = 0$$

$\therefore$ if $x^3 + qx + r = 0$ has roots α, β, γ then

$y^3 + \frac{q}{r}y^2 + \frac{1}{r} = 0$ has roots $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$

$\therefore$ coeffs are $1, \frac{q}{r}, 0, \frac{1}{r}$

or $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \gamma\alpha + \alpha\beta}{\alpha\beta\gamma} = -\frac{q}{r}$

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = 0$$

$$\frac{1}{\alpha\beta\gamma} = -\frac{1}{r}$$

etc.

c) from a)

$y^3 + 2qy^2 + q^2y - r^2 = 0$ has
roots $\lambda^2, \mu^2, \gamma^2$

Let $z = y^{-1}$ then

$$\frac{1}{z^3} + 2q\frac{1}{z^2} + q^2\frac{1}{z} - r^2 = 0$$

$$\text{i.e. } 1 + 2qz + q^2z^2 - r^2z^3 = 0$$

$$\text{i.e. } z^3 - \frac{q^2}{r^2}z^2 - \frac{2q}{r^2}z - \frac{1}{r^2} = 0$$

$\therefore$ coeffs of eqn with roots
 $\lambda^{-2}, \mu^{-2}, \gamma^{-2}$ are

~~$$1, -\frac{q^2}{r^2}, -\frac{2q}{r^2}, -\frac{1}{r^2}$$~~

1) For $n=1$

$$LHS = 1$$

$$RHS = 2 - \frac{1}{1}$$

$\therefore$ true for $n=1$

Assume true for $n=k$

$$\text{i.e. } 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} \leq 2 - \frac{1}{k}$$

Prove true for $n=k+1$

$$\text{i.e. } 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{k^2} + \frac{1}{(k+1)^2} \leq 2 - \frac{1}{k+1}$$

$$\text{Now } LHS \leq 2 - \frac{1}{k} + \frac{1}{(k+1)^2} \text{ by assumption}$$

$$= 2 - \left[\frac{(k+1)^2 - k}{k(k+1)^2} \right]$$

$$= 2 - \frac{k^2 + k + 1}{k(k+1)^2}$$

$$= 2 - \frac{1}{(k+1)} \times \frac{k^2 + k + 1}{k(k+1)}$$

$$= 2 - \frac{1}{k+1} \left[1 + \frac{1}{k^2 + k} \right]$$

$$< 2 - \frac{1}{k+1} \times 1$$

$$= 2 - \frac{1}{k+1}$$

$\therefore$ If it is true for $n=1$ it is true for $n=2$ and so on by induction for all n .

$$b) 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} < 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{99^2} \leq 2 - \frac{1}{99}$$

$$\therefore 1.46361 \leq 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{99^2} \leq 1.98$$

$$\text{but } 1.45 < 1.46361 \text{ and } 1.98 < 1.99$$

$$\therefore 1.45 < 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \frac{1}{99^2} < 1.99$$